Powers of Ten and Place Value

Multiplying and dividing by powers of ten, exponent form, and place-value shifts

Directions: Problem 1 shows a worked pattern to get you started. After that, work each problem on your own — watch whether you are multiplying or dividing, and by which power of ten.

1. Complete the pattern. The first line is done for you:

$7 \times 10 = 70$

$7 \times 100 =$ _________

$7 \times 1{,}000 =$ _________

2. $26 \times 10 =$ _________

3. $40 \times 1{,}000 =$ _________

4. $10^3 = 10 \times 10 \times 10 =$ _________

5. $3{,}500 \div 10 =$ _________

6. $62{,}000 \div 100 =$ _________

7. $45 \times 10^2 =$ _________

8. $8{,}000 \div 10^3 =$ _________

9. In 66,000, the 6 in the ten-thousands place is how many *times* the value of the 6 in the thousands place? _________
Explain how you know.

10. Compare the two 5s:
The digit 5 in 52,600 has a value of _________.
The digit 5 in 5,260 has a value of _________.
The first value is _________ times the second value.

11. How many times as large as 10^2 is 10^5?

Answer: ________________

12. One box holds 10 markers. One case holds 100 boxes. How many markers are in 6 cases?

Answer: ________________